

We select and review products independently. When you purchase through our links we may earn a commission. [Learn more.](#)

How-To Geek

What is the Difference Between 127.0.0.1 and 0.0.0.0?

AKEMI IWAYA [@asian_angel](#)

UPDATED AUG 16, 2015, 4:09 AM EST | 6 MIN READ



Most of us have heard of '127.0.0.1 and 0.0.0.0' but have probably not given much thought to them, but if both actually seem to point to the same location, then what is the actual difference between the two? Today's SuperUser Q&A post helps clear things up for a confused reader.

Today's Question & Answer session comes to us courtesy of SuperUser—a subdivision of Stack Exchange, a community-driven grouping of Q&A web sites.

Photo courtesy of [Kate Gardiner \(Flickr\)](#).

The Question

SuperUser reader Sagnik Sarkar wants to know what the difference between 127.0.0.1 and 0.0.0.0 is:

I understand that 127.0.0.1 points to *localhost* and that 0.0.0.0 also does as well (correct me if I am wrong). So, what is the difference between 127.0.0.1 and 0.0.0.0?

What is the difference between 127.0.0.1 and 0.0.0.0?

The Answer

SuperUser contributor DavidPostill has the answer for us:

What is the Difference Between 127.0.0.1 and 0.0.0.0?

- 127.0.0.1 is the loopback address (also known as *localhost*).
- 0.0.0.0 is a non-routable meta-address used to designate an invalid, unknown, or non-applicable target (a 'no particular address' place holder).

In the context of a route entry, it usually means the default route.

In the context of servers, 0.0.0.0 means *all IPv4 addresses on the local machine*. If a host has two IP addresses, 192.168.1.1 and 10.1.2.1, and a server running on the host listens on 0.0.0.0, it will be reachable at both of those IPs.

What is the IP Address 127.0.0.1?

127.0.0.1 is the loopback Internet protocol (IP) address also referred to as the *localhost*. The address is used to establish an IP connection to the same machine or computer being used by the end-user.

The same convention is defined for computers that support IPv6 addressing using the connotation of ::1. Establishing a connection using the address 127.0.0.1 is the most common practice; however, using any IP address in the range of 127...* will function in the same or a similar manner. The loopback

construct gives a computer or device capable of networking the capability to validate or establish the IP stack on the machine.

Source: [127.0.0.1 – What Are its Uses and Why is it Important?](#)

Special Addresses

The class A network number 127 is assigned the *loopback* function, that is, a datagram sent by a higher level protocol to a network 127 address should loop back inside the host. No datagram sent to a network 127 address should ever appear on any network anywhere.

Source: [Network Numbers](#)

If it is Whole Class A, What is the Point of Other Arbitrary Values for the Last Three Octets?

The purpose of the loopback range is testing of the TCP/IP protocol implementation on a host. Since the lower layers are short-circuited, sending to a loopback address allows the higher layers (IP and above) to be effectively tested without the chance of problems at the lower layers manifesting themselves. 127.0.0.1 is the address most commonly used for testing purposes.

Source: [IP Reserved, Loopback and Private Addresses](#)

For more information see the *Ask Ubuntu* question: [What is the Loopback Device and How do I Use it?](#)

What is the IP Address 0.0.0.0?

0.0.0.0 is a valid address syntax. So it should parse as valid wherever an IP address in traditional dotted-decimal notation is expected. Once parsed and converted to workable numeric form, then its value determines what happens next.

The all-zero value does have a special meaning. So it is *valid*, but has a meaning that may not be appropriate (and thus treated as not valid) for particular circumstances. It is basically the 'no particular address' placeholder. For things like address binding of network connections, the result can be to assign an appropriate interface address to the connection. If you are using it to configure an interface, it can instead remove an

address from the interface. It depends on the context of use to determine what 'no particular address' really does.

In the context of a route entry, it usually means the default route. That happens as a result more of the address mask, which selects the bits to compare. A mask of 0.0.0.0 selects no bits, so the comparison will always succeed. So when such a route is configured, there is always somewhere for packets to go (if configured with a valid destination).

In some cases, merely '0' will also work and have the same effect. But this is not guaranteed. The 0.0.0.0 form is the standard way to say 'no particular address' (in IPv6 that is ::0 or just ::).

Source: [What is the Meaning of the IP Address 0.0.0.0?](#)

In Internet Protocol version 4, the address 0.0.0.0 is a non-routable meta-address used to designate an invalid, unknown, or non applicable target. To give a special meaning to an otherwise invalid piece of data is an application of in-band signaling.

In the context of servers, 0.0.0.0 means *all IPv4 addresses on the local machine*. If a host has two IP addresses, 192.168.1.1 and 10.1.2.1, and a server running on the host listens on 0.0.0.0, it will be reachable at both of those IPs (*Note: This particular text is repeated from above as part of the overall answer*).

In the context of routing, 0.0.0.0 usually means the default route, i.e. the route which leads to 'the rest of' the Internet instead of somewhere on the local network.

Uses Include:

- The address a host claims as its own when it has not yet been assigned an address. Such as when sending the initial DHCPDISCOVER packet when using DHCP.
- The address a host assigns to itself when an address request via DHCP has failed, provided the host's IP stack supports this. This usage has been replaced with the APIPA mechanism in modern operating systems.

- A way to specify *any IPv4-host at all*. It is used in this way when specifying a default route.
- A way to explicitly specify that the target is unavailable.
Source: [127.0.0.1 – What Are its Uses and Why is it Important?](#)
- A way to specify *any IPv4 address at all*. It is used in this way when configuring servers (i.e. when binding listening sockets). This is known to TCP programmers as INADDR_ANY. [*bind(2) binds to addresses, not interfaces.*]

In IPv6, the all-zeros-address is written as ::

Source: [0.0.0.0](#) [Wikipedia]

DHCP Discovery/Request

When a client boots up for the first time, it is said to be in the *initializing state*, and transmits a DHCPDISCOVER message on its local physical subnet over User Datagram Protocol (UDP) port 67 (BootP server). Since the client has no way of knowing the subnet to which it belongs, the DHCPDISCOVER is an all subnets broadcast (destination IP address of 255.255.255.255), with a source IP address of 0.0.0.0. The source IP address is 0.0.0.0 since the client does not have a configured IP address.

If a DHCP server exists on this local subnet and is configured and operating correctly, the DHCP server will hear the broadcast and respond with a DHCPOFFER message. If a DHCP server does not exist on the local subnet, there must be a DHCP/BootP Relay Agent on this local subnet to forward the DHCPDISCOVER message to a subnet that contains a DHCP server.

This relay agent can either be a dedicated host (Microsoft Windows Server, for example) or a router (a Cisco router configured with interface level IP helper statements, for example).

...

After the client receives a DHCPOFFER, it responds with a DHCPREQUEST message, indicating its intent to accept the parameters in the DHCPOFFER, and moves into the *requesting*

state. The client may receive multiple DHCPOFFER messages, one from each DHCP server that received the original DHCPDISCOVER message. The client chooses one DHCPOFFER and responds to that DHCP server only, implicitly declining all other DHCPOFFER messages. The client identifies the selected server by populating the *Server Identifier* option field with the DHCP server's IP address.

The DHCPREQUEST is also a broadcast, so all DHCP servers that sent a DHCPOFFER will see the DHCPREQUEST, and each will know whether its DHCPOFFER was accepted or declined. Any additional configuration options that the client requires will be included in the options field of the DHCPREQUEST message. Even though the client has been offered an IP address, it will send the DHCPREQUEST message with a source IP address of 0.0.0.0. At this time, the client has not yet received verification that it is clear to use the IP address.

...

Client-server conversation for a client obtaining a DHCP address where the client and DHCP server reside on same subnet:

Packet Description	Source MAC Addr	Destination MAC Addr	Source IP Addr	Destination IP Addr
DHCPDISCOVER	Client	Broadcast	0.0.0.0	255.255.255.255
DHCPOFFER	DHCP Server	Broadcast	DHCP Server	255.255.255.255
DHCPREQUEST	Client	Broadcast	0.0.0.0	255.255.255.255
DHCPACK	DHCP Server	Broadcast	DHCP Server	255.255.255.255

Source: [Understanding and Troubleshooting DHCP in Catalyst Switch or Enterprise Networks](#)

Default Route

This document explains how to configure a default route or gateway of last resort. These IP commands are used:

- *ip default-gateway*
- *ip default-network*
- *ip route 0.0.0.0 0.0.0.0*

IP Route 0.0.0.0 0.0.0.0

Creating a static route to the network 0.0.0.0 0.0.0.0 is another way to set the gateway of last resort on a router. As with the *ip default-network* command, using the static route to 0.0.0.0 is not dependent on any routing protocols. However, IP routing must be enabled on the router.

Note: IGRP does not understand a route to 0.0.0.0. Therefore, it cannot propagate default routes created using the *ip route 0.0.0.0 0.0.0.0* command. Use the *ip default-network* command to have IGRP propagate a default route.

Source: [Configuring a Gateway of Last Resort Using IP Commands](#)

Have something to add to the explanation? Sound off in the comments. Want to read more answers from other tech-savvy Stack Exchange users? [Check out the full discussion thread here.](#)

AKEMI IWAYA

Akemi Iwaya has been part of the How-To Geek/LifeSavvy Media team since 2009. She has previously written under the pen name "Asian Angel" and was a Lifehacker intern before joining How-To Geek/LifeSavvy Media. She has been quoted as an authoritative source by ZDNet Worldwide. [READ FULL BIO »](#)

How-To Geek is where you turn when you want experts to explain technology. Since we launched in 2006, our articles have been read billions of times. [Want to know more?](#)